

Moein Razavi, Ph.D.

Senior AI/ML Engineer · Agentic AI & Multi-Agent Systems · Real-Time Decision Systems

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Professional Summary

Ph.D. Senior AI/ML Engineer specializing in **Agentic AI**, **multi-agent systems**, and **real-time decision systems** in production. **5+ years** architecting end-to-end GenAI platforms—from **LLM-orchestrated multi-agent pipelines** (LangGraph, RAG, vector DBs) to **transformer-based probabilistic forecasting** and **Monte Carlo decision engines** that act under uncertainty. At **Ford**, delivered a real-time **V2G aggregator/dispatch system**, a multi-agent EV analytics chatbot, and a multi-agent ingestion platform serving **10 US states** and **10+ TB** of telemetry. Known for shipping complete AI systems—LLM routing, probabilistic forecasting, uncertainty quantification, and real-time dispatch—rather than isolated models. Published researcher (**16 papers**, **470+ citations**, **h-index 10**), **4.0 GPA** across Ph.D./M.S. at Texas A&M.

Technical Skills

Agentic AI & Multi-Agent Systems: LangGraph · CrewAI · AutoGen · OpenAI Agents SDK · Semantic Kernel · MCP · LLM Routing & Tool Use · Planner/Executor & Supervisor/Worker patterns · ReAct · Self-Reflection · LangSmith Tracing & Eval

Generative AI & LLMs: GPT-4/4o · Claude · Gemini · Llama · Ollama · LangChain · Prompt Engineering · Fine-Tuning (LoRA/QLoRA) · Guardrails · Structured Outputs (Pydantic) · Function/Tool Calling · Streaming

RAG & Vector Databases: FAISS · ChromaDB · Pinecone · Hybrid Retrieval (BM25 + Dense) · Re-rankers · Chunking · Knowledge Graphs · Text-to-SQL · Document QA over PDF/Excel/HTML/XML

Decision Systems & Forecasting: Probabilistic Forecasting (Chronos-2, iTransformer, TFT, Autoformer) · Monte Carlo Simulation · P50/P90 Uncertainty Quantification · Real-Time Dispatch & Optimization · Bayesian Inference · Backtesting

ML / DL & MLOps: PyTorch · TensorFlow · Hugging Face Transformers · scikit-learn · XGBoost · Optuna · SHAP/LIME · Distributed GPU (Vertex AI) · Airflow · Docker · Kubernetes · Terraform · CI/CD (Tekton, GitHub Actions)

Cloud & Data: GCP (Vertex AI, BigQuery, Cloud Run, Dataproc, Looker) · AWS (S3, EC2, Lambda, SageMaker) · Azure (OpenAI) · PySpark · PostgreSQL · MongoDB · Redis · OAuth 2.0/JWT/SAML

Languages, Backend & Apps: Python (Expert) · TypeScript · SQL · Java · C++ · Scala · FastAPI · Flask · Django · Spring Boot · REST · WebSocket · gRPC · React · Next.js · Gradio · Streamlit · Swift/SwiftUI · Core ML

Professional Experience

Senior Data Scientist & AI/ML Engineer

Aug 2023 – Present

Ford Motor Company Detroit, MI

- **Architected production Multi-Agent GenAI Chatbot** (LangGraph, LangChain) with an **LLM Router/Decider layer** that dynamically dispatches user queries across specialized agents: **SQL Agent** (text-to-SQL over BigQuery with schema-aware prompting), **RAG Agent** (FAISS/ChromaDB vector store, hybrid retrieval, Chain-of-Thought reasoning, re-ranking), and **Document Agent**—serving EV analytics across **10 US states** and cutting end-to-end query latency by **35%**
- **Designed multi-agent data ingestion platform** (Vertex AI Gemini, LangGraph) composed of **Intake**, **Schema Discovery**, **Extraction**, **Validation**, and **Orchestrator** agents—replacing **~500 lines** of utility-specific parsers with a universal, self-healing pipeline that ingests CSV, Excel, PDF, XML, and HTML across heterogeneous utility partners at scale
- **Built real-time V2G (Vehicle-to-Grid) aggregator and dispatch system** combining **transformer-based probabilistic forecasting** (Chronos-2, iTransformer) with **Monte Carlo simulation** (**~3,000 scenarios per dispatch**) to optimize bidding and energy dispatch under uncertainty; productionized with **P50/P90 quantile reporting**, sub-second decision latency, and streaming telemetry feedback loops
- **Engineered the underlying transformer forecasting pipeline** (TFT, Autoformer, Informer) with Hugging Face and Optuna-driven hyperparameter search on distributed Vertex AI GPU—improved EV load & demand forecasts **10%** and cut training time **40%**; outputs feed rate optimization, dispatch, and capacity planning
- **Built XGBoost-based decision model** for EV energy bill prediction over vehicle telematics, utility tariffs, and home consumption ($R^2 = 0.85$); integrated into the rate recommender as a real-time inference service with backtesting and drift monitoring
- **Spearheaded Rate Recommendation MVP** end-to-end (Flask, BigQuery, Dash with SAML/ADFS, Cloud Run, Tekton CI/CD)—a production decision system shortening the rate-design iteration cycle by **20%**
- **Engineered 10+ TB EV telemetry pipelines** with PySpark, BigQuery, and Airflow DAGs powering all downstream agents and forecasting models; Looker Studio dashboards automated KPI reporting for **500+ stakeholders**, shrinking turnaround **45%**
- **Deployed production MLOps infrastructure** on GCP (Cloud Run, Vertex AI, Dataproc) with Tekton, Terraform IaC, SonarQube, and 42 Crunch security—achieving **99.9% uptime** and reducing ingestion time by **25%**

Software Engineer & AI/ML Researcher

CereVu Medical, Inc. San Francisco Bay Area, CA

- **Architected end-to-end real-time biosignal AI platform** streaming **9 biomarkers** from wearable patches through Python microservices to clinician apps with embedded **ROPA pain decision algorithm**—elevating monitoring precision **30%**
- **Deployed LSTM/CNN models on-device** via **Core ML** for real-time inference on wearable patches across **400+ patients** in 5 clinical studies—**15% accuracy lift**, real-time video at **60 FPS**
- **Built scalable Python microservices** (FastAPI, Flask) with WebSocket biosignal streaming handling **10K+ concurrent connections** (**40%** latency reduction); powered a **React/Next.js clinical decision dashboard** (MongoDB, JWT) for physician workflows
- **Shipped 2 production SwiftUI iOS apps** (Vitality + Clinical) with CoreBluetooth (Nordic UART BLE), role-based MVVM-Coordinator architecture, and OAuth 2.0/JWT; owned CI/CD (Tekton, AWS CodeDeploy, Docker) for **30% faster** zero-downtime releases

Jan 2023 – Present

Founder & Lead Developer

Clink Social | Inspecalytics McKinney, TX

- **Architected Clink Social** end-to-end on **Firebase** (Auth, Firestore, Storage, FCM) with real-time messaging, fan-out timeline, and secure **QR-based cross-device OAuth/JWT login**—boosted engagement **22%**, cut sign-in friction **35%**
- **Built Inspecalytics** field-inspection suite on **Supabase** with offline-first SHA-256 sync, in-app DOCX/PPTX/XLSX report generation, and Fastlane + GitHub Actions CI/CD—reduced report cycles **35%**, deployment time **40%**

Sep 2018 – May 2024

Research Assistant & ML Engineer

Texas A&M University College Station, TX

- **Designed dual-branch transformer** (1D-CNN tokenizer, RoPE, cross-attention, focal loss) for real-time wearable stress detection on **noisy, naturalistic Apple Watch HR data (256 participants, 1.7B rows)**; **AUC 0.993, F1 0.965**, deployed on-device via **Core ML** at **60% faster inference**
- **Deployed real-time computer vision system** for COVID-19 distancing/mask detection in production environments, **98% accuracy** across **50K+ daily interactions (76 citations)**
- **Built OpenSync**—open-source streaming platform synchronizing EEG, eye-tracking, and biosensors via WebSocket pipelines on noisy multi-modal data; adopted by **3 research labs (31 citations, J. Neuroscience Methods)**

Education

Ph.D., Industrial Engineering (Data Science Concentration) | GPA: 4.0/4.0 | Dean's List | Graduate Research Fellowship Aug 2021 – Dec 2023

Texas A&M University, College Station, TX

Dissertation: Advancing Stress Detection—Real-Time, Naturalistic, and Personalized AI Approaches for Mental Disorder Detection.

Aug 2019 – May 2021

M.S., Computer Science | GPA: 4.0/4.0 | Dean's List | Graduate Research Fellowship

Texas A&M University, College Station, TX

Coursework: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Algorithm Design, Database Systems, Software Engineering.

Selected Publications (16 papers · 470+ citations · h-index 10)

Razavi, M., et al. "ML, DL & Data Preprocessing for Stress & Mental Disorder Detection: Scoping Review." *JMIR Mental Health*, 2024. [79 cites]

Kazeminasab, S., . . . , **Razavi, M.**, et al. "Localization, Mapping & Navigation in In-Pipe Robots: A Review." *IEEE Access*, 2021. [85 cites]

Razavi, M., et al. "Automatic Monitoring of Physical Distance & Mask Compliance in COVID-19." *SN Computer Science*, 2022. [76 cites]

Razavi, M., et al. "OpenSync: Synchronizing Multi-Modal Measures in Neuroscience." *J. Neuroscience Methods*, 2022. [31 cites]

Razavi, M., McDonald, A., Mehta, R., Sasangohar, F. "Evaluating Mental Stress via Wearable HR & Hand Acceleration." *arXiv:2309.11097*, 2023. [26 cites]

Certifications & Honors

Generative AI with LLMs – AWS · **Deep Learning Specialization** – DeepLearning.AI · **Machine Learning** – Stanford · **NLP Specialization** – DeepLearning.AI · **Advanced MLOps** – Coursera · **Bayesian Statistics** – Coursera · **Graduate Research Fellowship**, Texas A&M (2019–2023)